

PyCCEA: A Python package of cooperative co-evolutionary algorithms for feature selection in high-dimensional data

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

Feature selection is a critical preprocessing step in many machine learning pipelines, particularly when dealing with high-dimensional datasets commonly found in domains such as genomics, text mining, and image analysis. However, many feature selection techniques, such as heuristic search methods and evolutionary algorithms, struggle to maintain predictive performance and interpretability as the dimensionality of data increases (?).

Cooperative co-evolutionary algorithms (CCEAs) offer a promising approach for tackling this challenge by dividing the high-dimensional space into multiple tractable low-dimensional subcomponents. Each subcomponent is evolved independently using a subpopulation, and candidate solutions are evaluated based on their collaboration with representatives from other subpopulations (Ma et al., 2018).

PyCCEA implements CCEAs specifically for feature selection in high-dimensional data. It provides modular components for decomposition, collaboration, fitness evaluation, and optimization, allowing users to reproduce existing methods or develop new ones. The framework currently only follows a wrapper-based approach, in which subsets of features are evaluated using machine learning models to guide the search. However, this design is not restrictive, PyCCEA can be extended to support other evaluation paradigms, such as filters, hybrid, or embedded methods, which are planned as future features.

All machine learning and data processing components are built on top of well-established libraries such as scikit-learn and Pandas, ensuring compatibility, flexibility, and ease of use. To the best of our knowledge, PyCCEA is the first open-source package dedicated to CCEAs for feature selection, offering a standardized and extensible framework for tackling high-dimensional datasets in a reproducible manner.

Statement of need

Despite the growing interest in cooperative co-evolutionary algorithms for feature selection and their promising results (Song et al., 2020), there is currently no publicly available software package that consolidates these techniques into a reusable and extensible tool. Existing research typically relies on custom [Rashid et al. (2020); (?); Firouznia:2023] or unpublished implementations (Song et al., 2020; ?), which makes it difficult to reproduce results, compare methods, or build upon previous work.

PyCCEA addresses this gap by providing a well-organized, research-focused implementation of

40 cooperative co-evolutionary algorithms tailored to feature selection. It incorporates widely
41 used strategies from the literature and encourages standardization in experimental design and
42 evaluation. By enabling consistent benchmarking and facilitating the development of new
43 strategies, PyCCEA supports both methodological innovation and practical application in high-
44 dimensional machine learning problems. Its release lowers the barrier to entry for researchers
45 and practitioners, accelerating progress in the field of feature selection using evolutionary
46 computation.

47 Recent research in evolutionary computation and feature selection has already leveraged PyCCEA
48 as a foundational framework for implementing and evaluating cooperative co-evolutionary
49 strategies (Venâncio & Batista, 2025), demonstrating its value in supporting reproducible
50 and extensible scientific contributions. Future improvements aim to broaden its evaluation
51 capabilities and encourage further community-driven development.

52 Acknowledgements

53 This work was supported by Brazilian agencies FAPEMIG (Research Support Foundation of
54 the State of Minas Gerais), CNPq (The National Council for Scientific and Technological
55 Development), CAPES (Coordination for the Improvement of Higher Education Personnel) –
56 Finance Code 001, and the Operations Research and Complex Systems Laboratory (ORCS
57 Lab./UFMG). Professor Lucas S. Batista is a FAPEMIG-CNPQ scholarship holder (process
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